- Why It Matters: When a user tries to log into the network, the Global Catalog helps find their account details across the whole forest super fast. Even if the first DC is offline, it allows users to log in smoothly by verifying things like Universal Group memberships.
- For the Lab: This option must always be checked (✓).

3. Read-Only Domain Controller (RODC)

- In Simple Terms: This is a special type of Domain Controller where the Active Directory database is "Read-Only," meaning data can be read but cannot be modified directly on this server.
- Why It Matters: It is heavily used in low-security branch offices. If a malicious user physically steals this server from that remote office, they cannot modify or damage the master Active Directory database because this server is blocked from sending updates (writing) back to the main DC.
- For the Lab: Since we are building a full-privilege Additional DC, this option must be left unchecked (empty).



Here we want to set the restore password. The DSRM password is not used for daily administrative logons.

- Standard Logon: To log into the server normally, you must continue to use your standard Domain Administrator credentials.
- DSRM Logon Scope: The DSRM password is strictly reserved for emergency recovery states. It becomes active only when the server is booted into Directory Services Restore Mode (Safe Mode) to fix a corrupted or broken Active Directory database.

Then click Next.

Click Next on the DNS option.

Replication

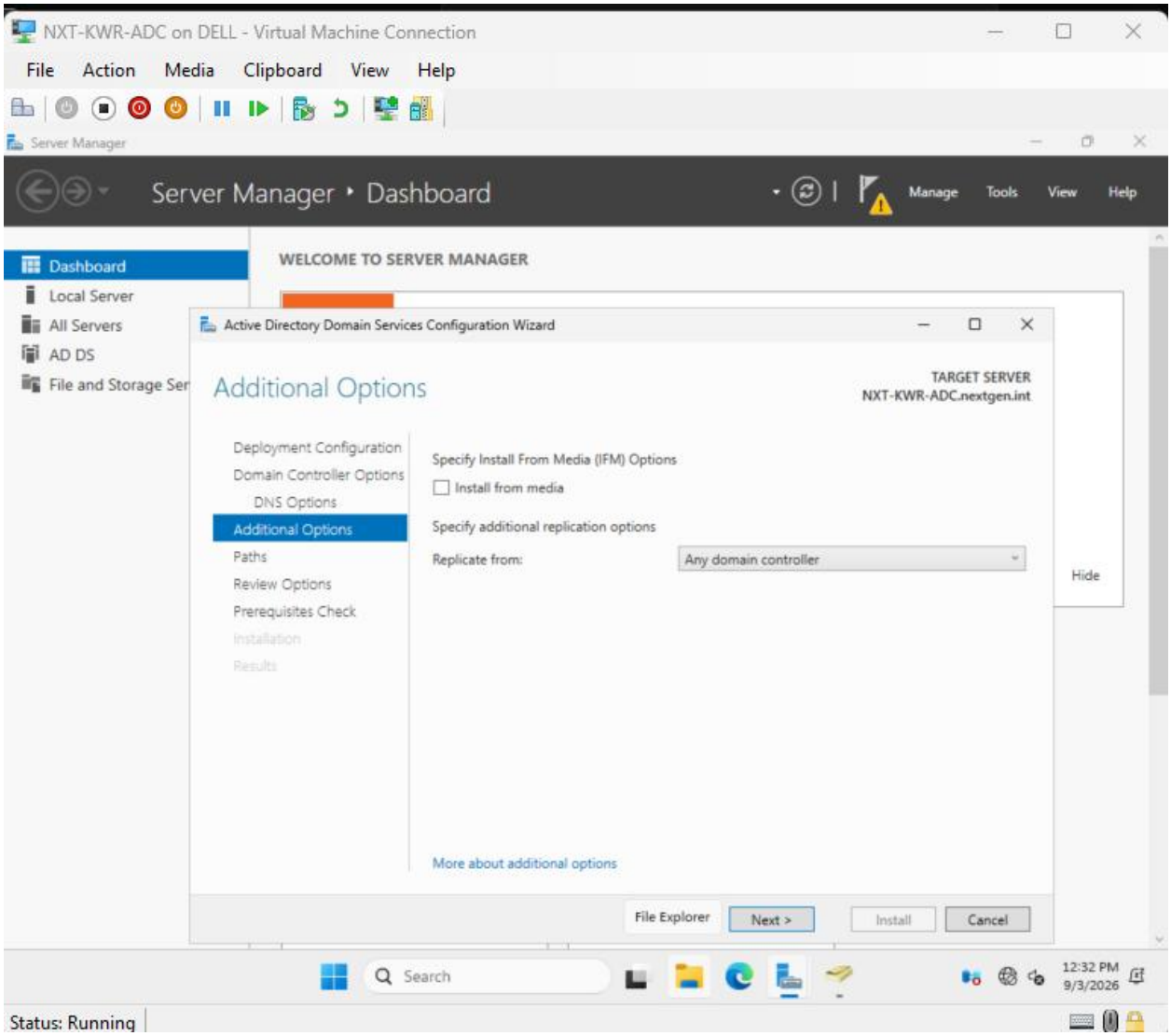
What is Active Directory Replication?

Active Directory Replication is the automatic process by which changes (such as creating users, modifying passwords, or updating group policies) made on one Domain Controller are copied and synchronized to all other Domain Controllers within the network.

- Core Mechanism: It ensures that every Domain Controller hosts an identical, up-to-date copy of the Active Directory database (ntds.dit).
- Multi-Master Model: Active Directory uses a multi-master replication model, meaning you can add or change data on any Domain Controller, and those updates will automatically replicate to the rest of the servers.

Primary Benefits:

- High Availability / Redundancy: If one server crashes, other servers can handle user authentication because they contain the exact same replicated database.
- Load Balancing: Distributes the network logon traffic across multiple domain controllers to improve performance.



What is the "Replicate from" option?

The "Replicate from" option specifies which existing Domain Controller (DC) the new Additional Domain Controller (ADC) should use to copy (replicate) the initial Active Directory database (ntds.dit).

- Any Domain Controller (Default): The system automatically finds the closest and fastest available Domain Controller in the network to pull the configuration and user data.
- Specific Domain Controller: Allows the administrator to manually pick a specific server (e.g., the Primary DC) to source the data, which is useful in large networks with multiple sites to control network traffic.

Choose any domain controller.

Then click Next, and after all the remaining options, keep the defaults and click Next, then install the server.

Very important

After promoting, when I log in to the ADC the password was wrong — why?

Why Did the "Incorrect Password" Error Happen?

The problem was not actually a wrong password, but the username format you used to log in.

- Before Promotion: The server was just a standard computer (Workgroup PC). You logged into it using its own Local Administrator account.
- After Promotion: Now that the server is successfully promoted to an ADC, all its old local accounts are disabled. It is now a core part of the domain structure. Therefore, you must log in using the Domain Administrator credentials from your Main DC (VM 1).

How to Properly Log into the ADC Screen

- Go to the server's Windows Login Screen.
- If it displays a default name like Administrator, click on "Other User" in the bottom-left corner.
- In the Username field, type your domain name followed by the username:

nextgen-ad\Administrator (or Administrator@nextgen.int)

- In the Password field, enter the exact Domain Administrator password that you use on your Main DC (VM 1).

Why Does It Feel Like You Logged into the Existing Main DC?

When you log in using nextgen-ad\Administrator, the desktop profile looks identical to your Main DC. This makes it feel like you connected back to VM 1.

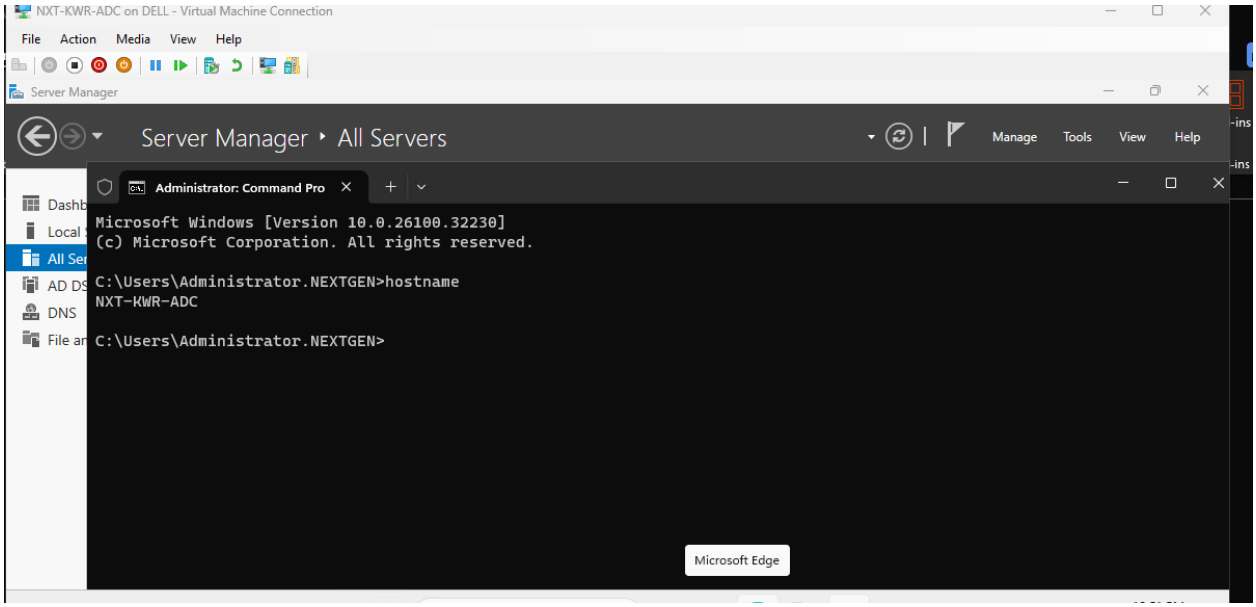
- The Reality: Because this server is now an ADC (Additional Domain Controller), it holds an exact replica (copy) of the Main DC's database. The nextgen-ad\Administrator account exists on both servers simultaneously.
- When you type that name on the ADC monitor, you are opening the ADC's own physical desktop, not the Main DC's desktop.

How to Verify You Are 100% on the ADC Machine

To prove to yourself (or your instructor) that you are physically operating inside the ADC machine, perform this quick verification check after logging in:

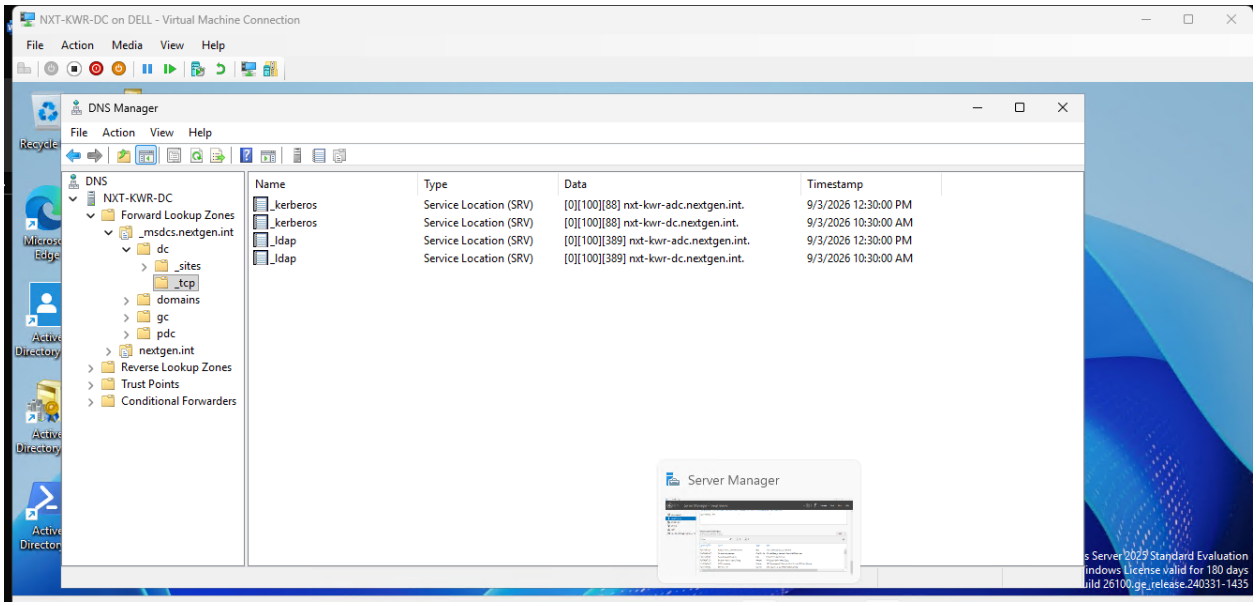
- Open the Command Prompt (CMD) on the desktop.
- Type hostname and press Enter.
- The screen will output the unique computer name of your ADC server (for example, a random string like WIN-27M6SLPLCOD).

If you see the ADC's computer name instead of your Main DC's name, it confirms you are working on the correct machine!



"After promoting VM2 as an Additional Domain Controller, the local user account (nextgen-adc) is no longer used for login. The server now uses Active Directory domain accounts. Therefore, I must log in to both the Main Domain Controller (VM1) and the Additional Domain Controller (VM2) using the same domain user account (nextgen\nextgen-ad). The only difference is the computer I am logging into."

Let's check whether the DC applies as ADC correctly:



Now let's check whether the ADC server works properly.

Firstly, open CMD with Run as admin.

1) Check Sync and Replication

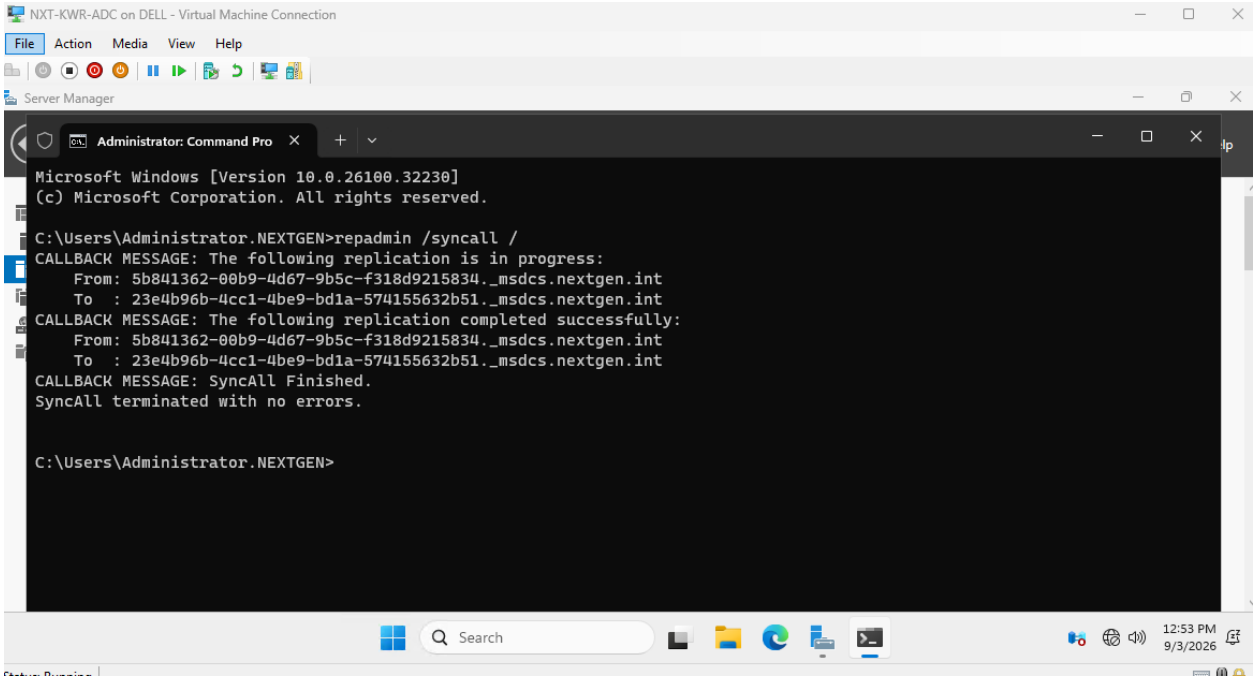
Command: repadmin /syncall /

1. What Is Replication?

- Definition: Replication is the automated background process of copying and synchronizing Active Directory data (such as new users, passwords, groups, and policies) between all Domain Controllers (DCs).

2. The repadmin /syncall Command

- What it does: This command is used by administrators to manually force immediate Active Directory replication across all Domain Controllers.

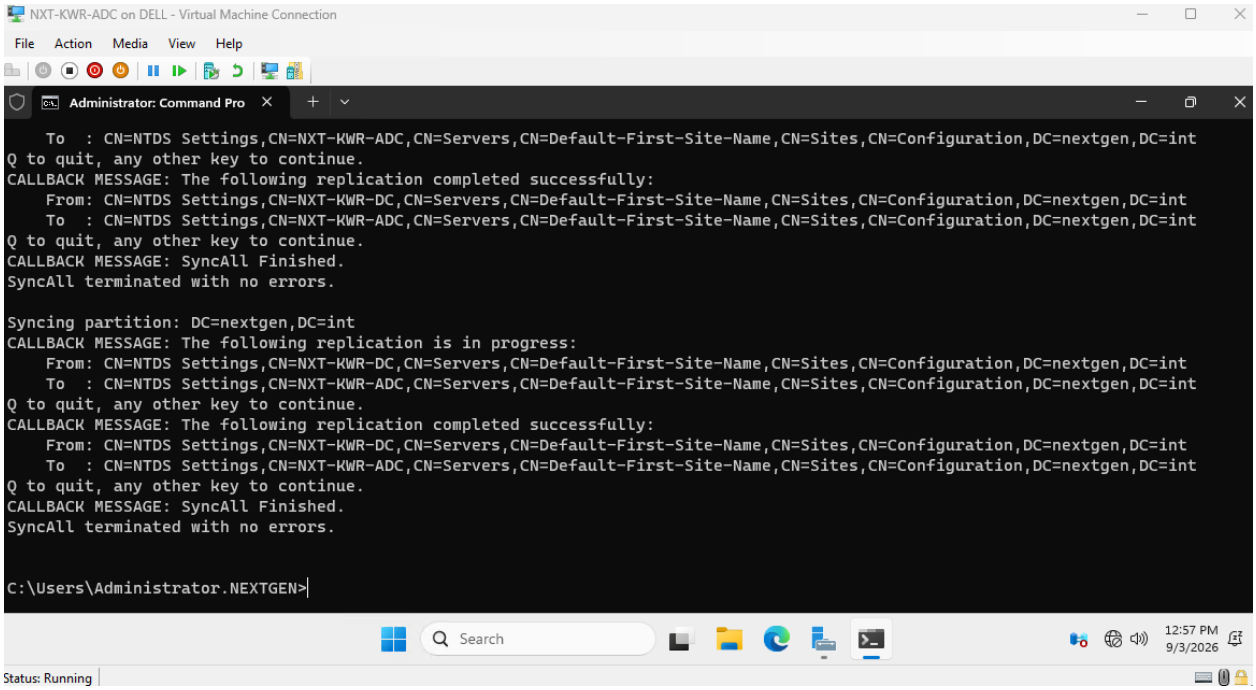


We choose only the A, d, e, P options in this situation. If you want to check others you can use these commands:

repadmin /syncall /Adep

> 💡 *Each letter is a switch: A = all partitions, d = display names, e = ignore site boundaries (sync across sites too), P = push changes outward. Combining them forces a full, immediate replication in every direction.*

When we run this command on the ADC, the Main DC must be switched on.

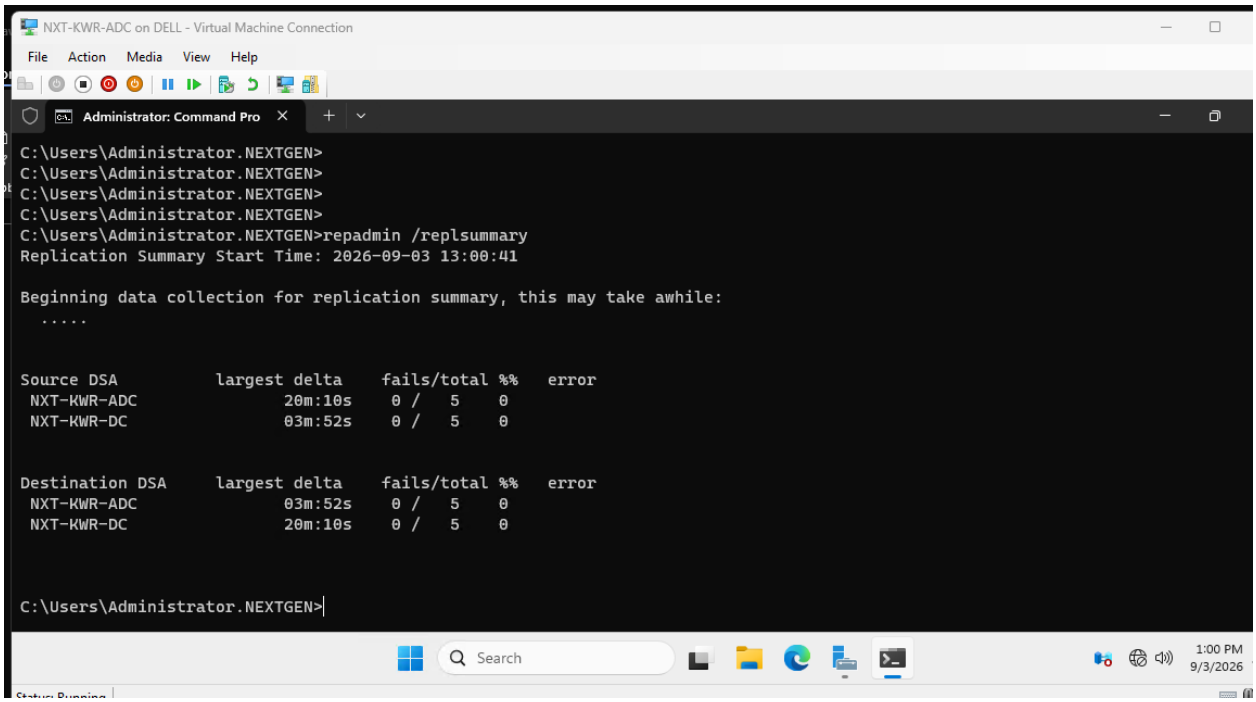


Here, replication completed successfully.

2) Check DNS or Replication Summary

repadmin /replsummary

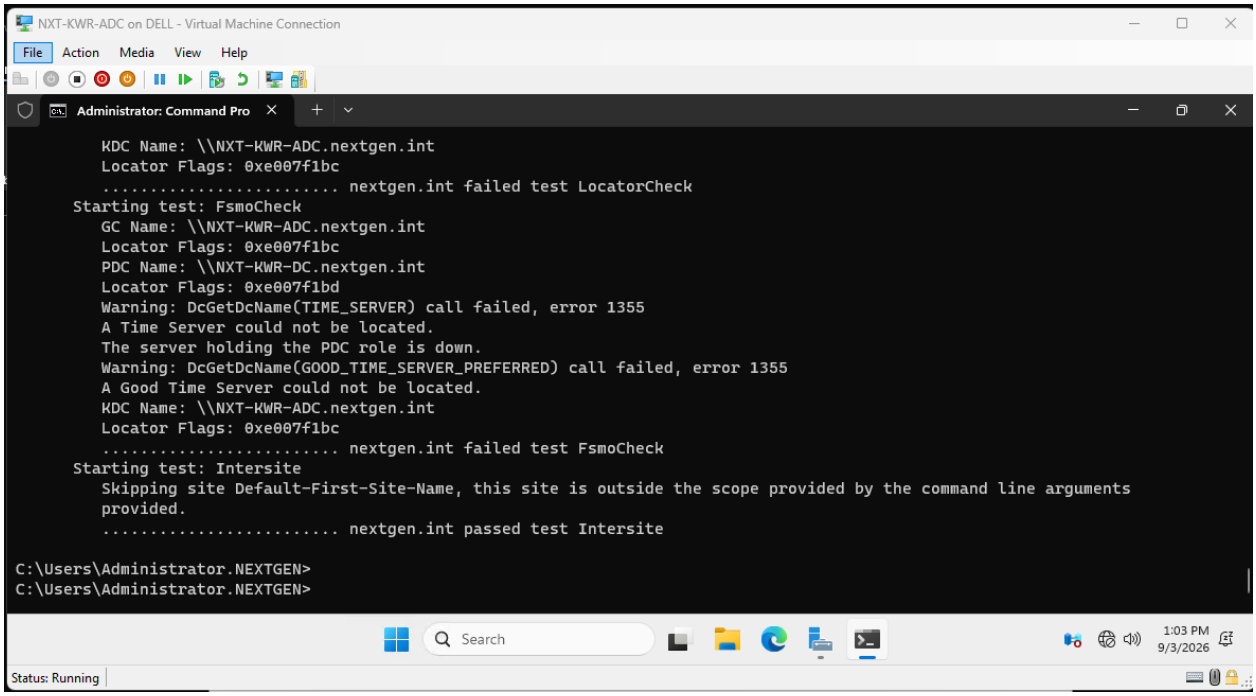
I got some error:



Working properly, no errors.

3) Let's Check the Health of the DC

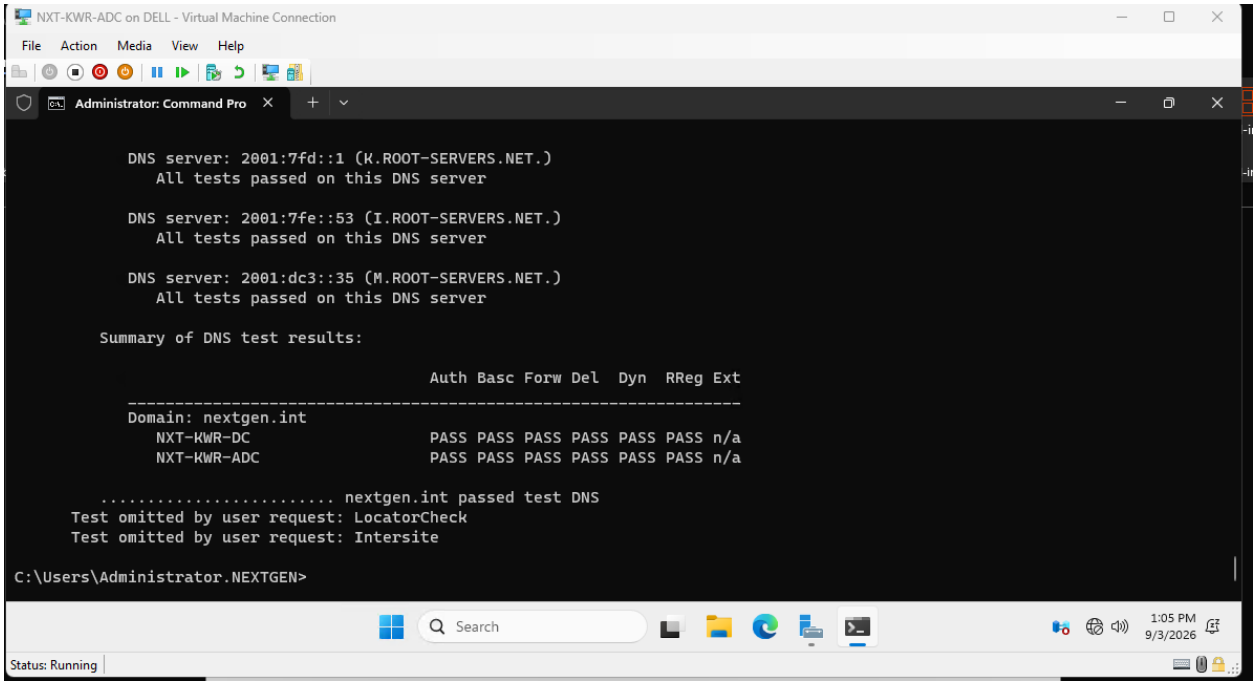
dcdiag /e /c /v



This should pass all tests.

4) Check All DNS Records / Check DNS Resolving

dcdiag /e /test:dns /v



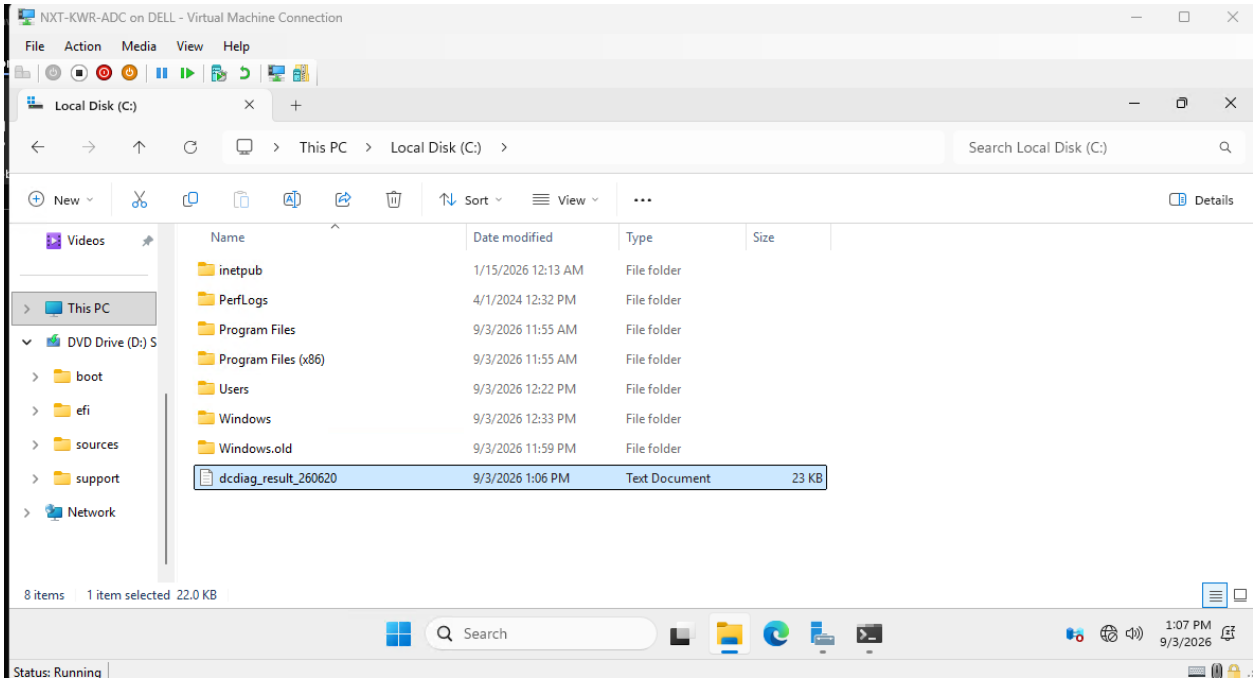
5) Save Any Result as a Text File

Example: let's save the result of dcdiag /e /test:dns /v as a .txt file.

dcdiag /e /test:dns /v > c:\dcdiag_result_260620.txt

{command > folderpath\filename.extension}

> *The > symbol redirects the command's output into a file instead of showing it on screen — useful for attaching proof of your health checks in the report.*



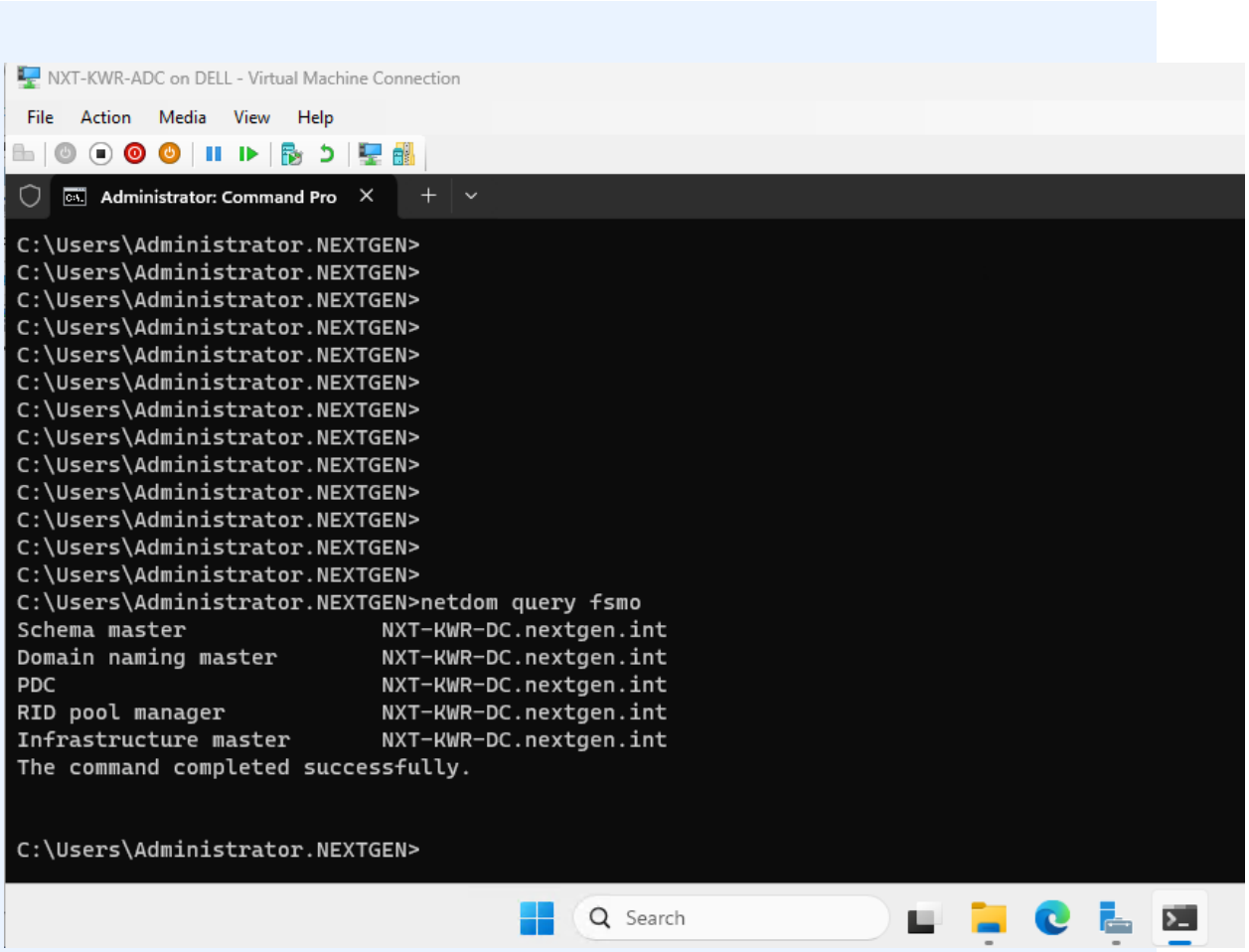
If we enter switches, we must put a / in front of each switch — this is important. Example: /e /test:dns /v

6) netdom Queries

netdom query /?

Now you can see all the available queries and check any one of them with this command.

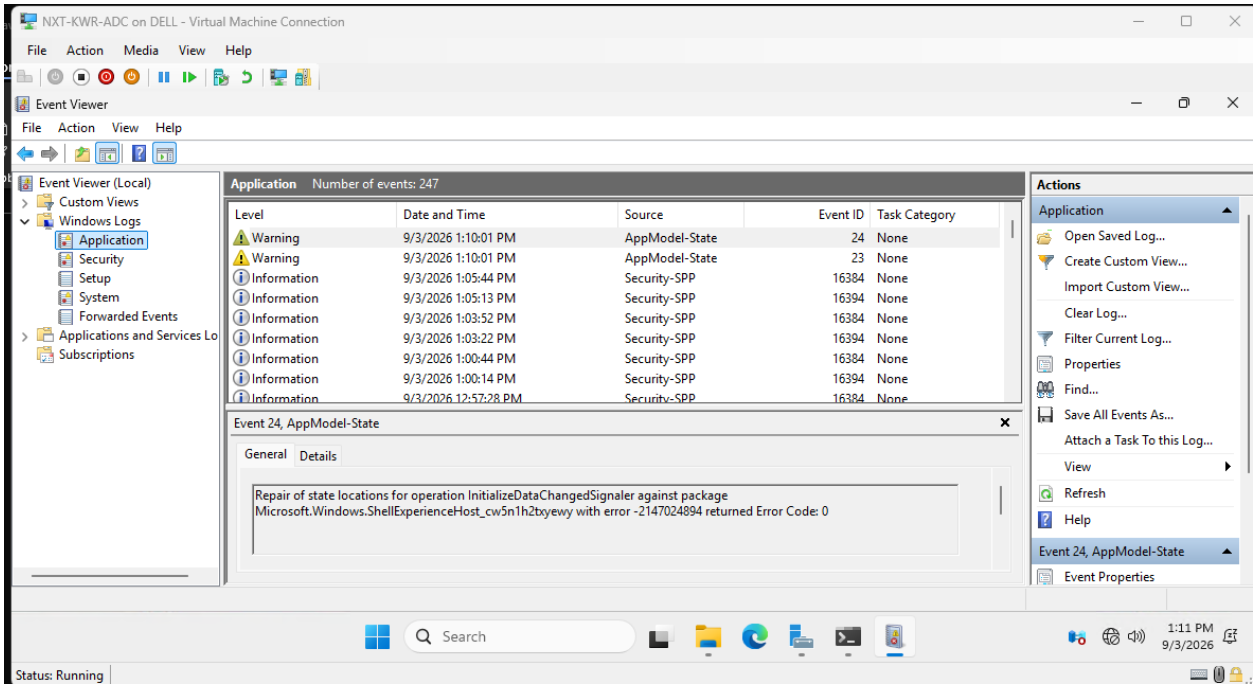
Let's check netdom query fsmo (FSMO roles):



FSMO (Flexible Single Master Operations) roles are special jobs that only one DC in the forest/domain can hold at a time, e.g. Schema Master, Domain Naming Master, PDC Emulator. This query shows you which server currently holds each role.

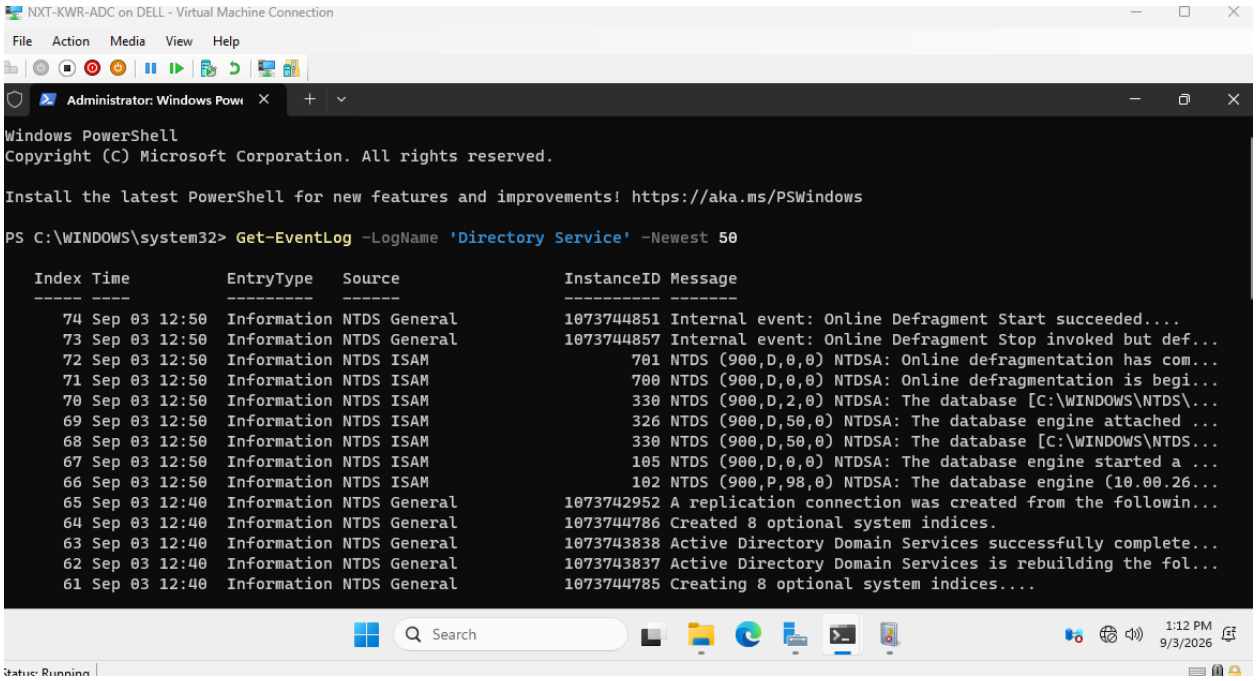
7) Check All Logs

We can check all logs via Event Viewer.



Also, we can do this with PowerShell commands — we can get all the commands from the internet. Example: we can check Directory Services issues, and we can check replication issues.

[PowerShell output of Get-EventLog -LogName 'Directory Service' -Newest 50]



Get-EventLog: Specifies the core PowerShell cmdlet designed to read localized system event logs.

- -LogName "Directory Service": Filters the query specifically to target logs generated by the Active Directory Domain Services (AD DS) engine.

- -Newest 50: Restricts the console output to display only the 50 most recently recorded logs, making it highly efficient for scanning active errors.